

Revision Notes for Class 7 Science Chapter 8 – Measurement of Time and Motion

People have always been interested in keeping track of time. They noticed patterns in nature, such as the rising and setting of the Sun or the changing seasons, which repeat after fixed intervals. Early humans began to use these recurring events to divide time into days, months, and years. As daily life became more complex, people needed ways to measure smaller intervals during a day.

Measurement of Time in Ancient Days

To tell time before clocks were invented, several creative tools were developed. Sundials measured time by tracking the Sun's shadow on marked surfaces. Water clocks used either the steady flow of water out of a vessel with time markings, or a floating bowl with a small hole that slowly filled and sank at intervals. Hourglasses counted time based on sand moving from one chamber to another. Candle clocks used burning candles with marks to indicate how much time had passed.

Interesting Facts About Historical Clocks

The Samrat Yantra at Jantar Mantar in Jaipur is the world's largest sundial, and its shadow moves about 1 millimetre every second. In India, time was also measured using water clocks such as the Ghatika-yantra, described in ancient texts by scholars like Aryabhata. The Ghatika-yantra played a vital role in religious and public activities, and the passing of time was often announced by sounds from drums or conches.

The Pendulum and Mechanical Timekeeping

By the 17th century, precise timekeeping became essential as travel increased. The pendulum clock was a breakthrough. Christiaan Huygens invented and patented the pendulum clock in 1656, inspired by Galileo's observations. He discovered that the time taken for a pendulum to swing depends only on its length, not on the mass of its bob.

A simple pendulum is a metal ball (bob) hung from a thread fixed to a rigid support. When set swinging, it moves in periodic motion. One complete to-and-fro movement is called one oscillation, and the time for this is the "time period." The time period depends on the length of the pendulum. If we measure how long the pendulum takes to complete 10 oscillations and divide that by 10, we get the time for one oscillation.

Modern Clocks and Accurate Measurement

All clocks, from the oldest to the newest, work by marking equal intervals using some repetitive process. Quartz clocks use the vibration of special crystals. Atomic clocks use the repeating vibration of atoms and are accurate to just one second lost in millions of years. This level of precision is important for science, medicine, sports, and technology.

The SI unit for time is the second (s). Larger units include the minute (min) and hour (h), remembering that 60 seconds make one minute and 60 minutes make one hour. Always write unit symbols in lowercase and singular (e.g., 6 min, 3 h). Leave a space between the number and unit.

Speed: Understanding Fast and Slow

When we say something is fast or slow, we are comparing how much distance it covers in a certain amount of time. In a race, the fastest person is the one who covers the most ground in the least time. Speed shows how quickly an object moves.

Speed is calculated using the formula: **Speed = Total distance covered / Total time taken**. The SI unit is metre per second (m/s), but kilometres per hour (km/h) is also common, especially for longer distances or vehicles.

For example, if Swati cycles 3.6 km in 15 minutes to school, her speed is calculated by:
 $3.6 \text{ km} \times 1000 = 3600 \text{ m}$; $15 \text{ min} \times 60 = 900 \text{ seconds}$. $\text{Speed} = 3600 \text{ m} / 900 \text{ s} = 4 \text{ m/s}$.

Relationship Between Speed, Distance, and Time

If you know any two of the following — distance, time, or speed — you can find the third using these relations:

- **Speed = Distance / Time**
- **Distance = Speed × Time**
- **Time = Distance / Speed**

Uniform and Non-uniform Linear Motion

If an object moves along a straight path (linear motion) and covers equal distances in equal time intervals, its motion is called uniform linear motion. For example, if a train moves 20 km every 10 minutes, its speed is uniform. If the distance covered in equal time intervals varies, it is non-uniform motion—common in daily life. Often, vehicles speed up or slow down due to traffic or road conditions, so the speed is not constant, and we talk about average speed.

The movement of two trains over time can be observed in a table:

Distances travelled by two trains in equal 10 min intervals

Time (AM)	Position (km) Train X	Distance (km)	Position (km) Train Y	Distance (km)
10:00	0	0	0	0
10:10	20	20	20	20
10:20	40	20	35	15
10:30	60	20	50	15
10:40	80	20	75	25
10:50	100	20	95	20

11:00	120	20	120	25
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Daily Life Connections and Interesting Uses

Modern clocks help us to measure not just hours and minutes, but even fractions of a second. This is very important in fields like sports, where races are often decided by one-hundredth of a second. In vehicles, speedometers display current speed, while odometers measure distance travelled. In technology and medicine, accurate timekeeping is essential for proper working of devices like smartphones and heart monitors.

To sum up, the time taken by a pendulum for one complete movement is called its time period, and the time period depends mainly on the length of the pendulum. The SI unit for time is the second. Speed is the distance covered in a given time, and uniform motion means equal distances in equal times, whereas in non-uniform motion, the speed keeps changing. Most things in real life do not move at uniform speed.

- Timekeeping started with observing nature's regular events and developed into complex devices like sundials, water clocks, and eventually, atomic clocks.
- A simple pendulum's time period depends mainly on its length and is used to measure time.
- Speed is a measure of how fast an object moves; calculated with the formula:
 $\text{Speed} = \text{Distance} / \text{Time}$.
- Uniform motion is when an object covers equal distances in equal times; non-uniform motion is more common in everyday life.
- Modern measuring devices allow for extreme precision, important in sports, technology, and scientific exploration.